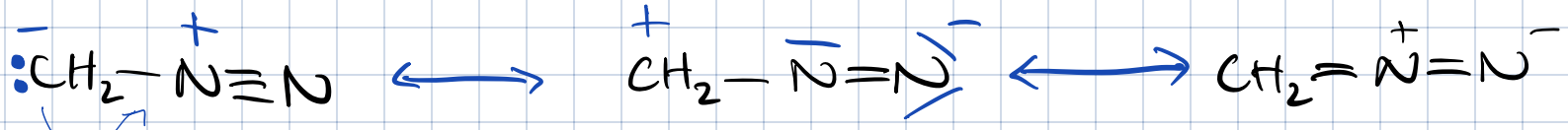
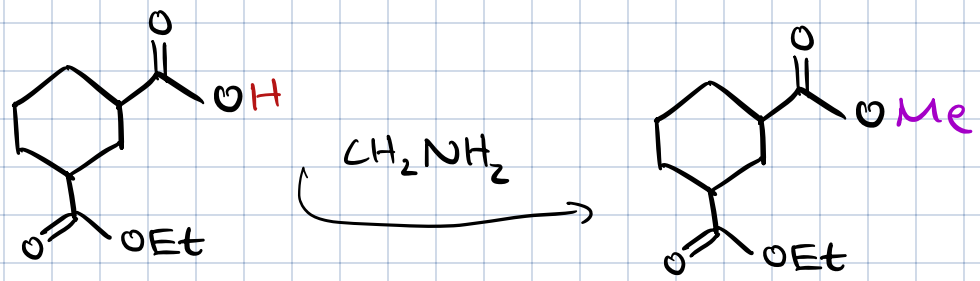
Diazomethane



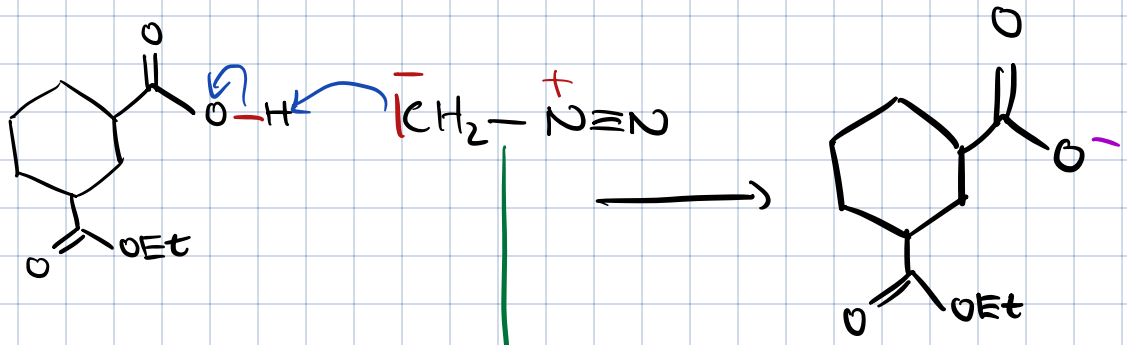
Use:

Esterification of carboxlic acids while avoiding transesterification

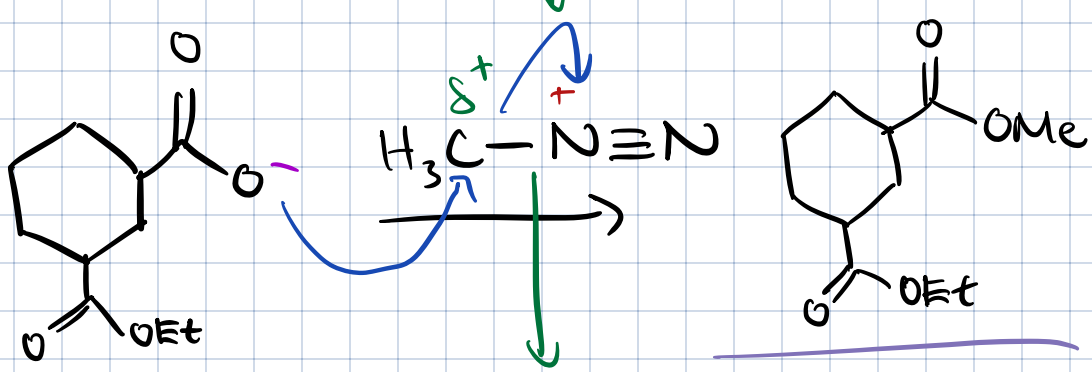


Mech:

1. Protonation



1. SN2



$N_2(g)$

Unlike Fischer, CH_2N_2 doesn't need acid, and thus avoids rxn's involving the pre-existing ester group.

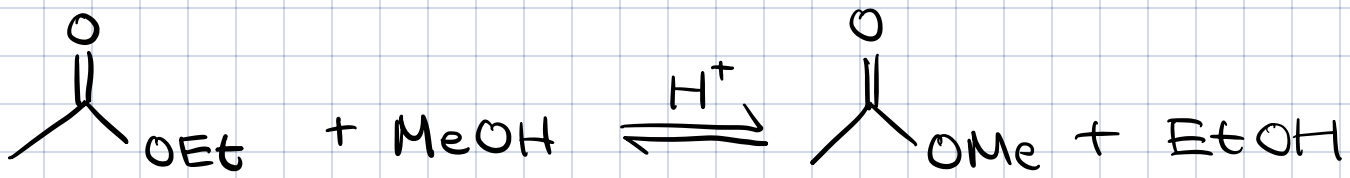
★ A 2-step alternative would be forming a acyl chloride, which will esterify w/o H^+

1. $SOCl_2$
2. MeOH

- Notice how protonation of the reagent by the substrate starts the rxn, making CH_2N_2 CA-selective

- **Really just for methyl esters** (unstable $RCHNH_2$ & SN2)

Transesterification



same mech as Fischer

Takeaway:

Ester + ROH + H^+ will equillibriate a transesterification